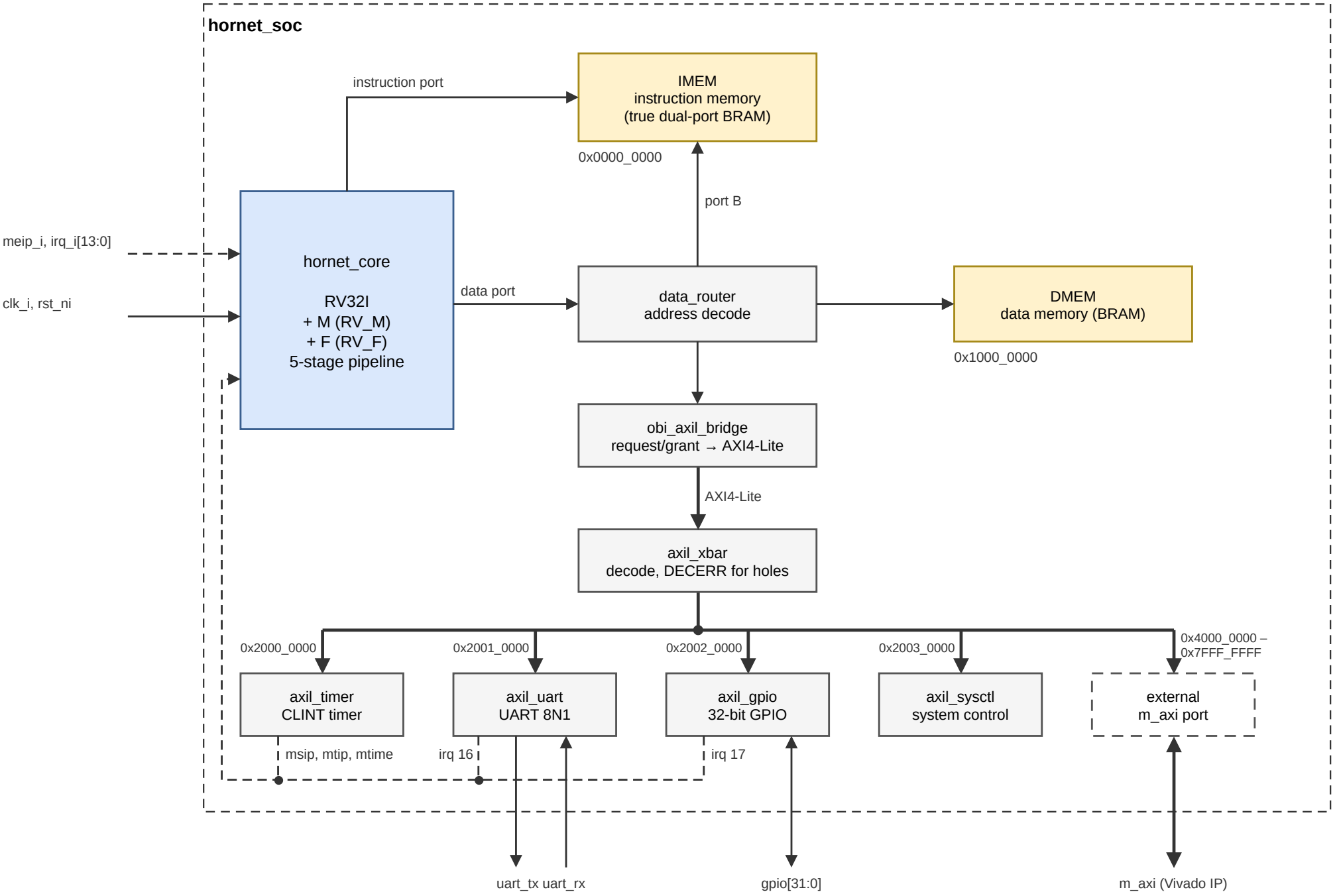
hornet_soc
instruction port
IMEM
instruction memory
(true dual-port BRAM)
0x0000_0000
port B
meip_i, irq_i[13:0]
clk_i, rst_ni
hornet_core
RV32I
+ M (RV_M)
+ F (RV_F)
5-stage pipeline
data port
data_router
address decode
DMEM
data memory (BRAM)
0x1000_0000
obi_axil_bridge
request/grant → AXI4-Lite
AXI4-Lite
axil_xbar
decode, DECERR for holes
0x2000_0000
0x2001_0000
0x2002_0000
0x2003_0000
0x4000_0000 –
0x7FFF_FFFF
axil_timer
CLINT timer
axil_uart
UART 8N1
axil_gpio
32-bit GPIO
axil_sysctl
system control
external
m_axi port
msip, mtip, mtime
irq 16
irq 17
uart_tx uart_rx
gpio[31:0]
m_axi (Vivado IP)